

Technical Challenge: Logistics Decision-Intelligence Prototype

Objective Build a functional data pipeline that integrates internal and external variables to generate a Prescriptive Operational Decision for a regional delivery hub manager.

1. The Dataset (Simplified) You are provided with (or should mock) the following data points:

- Orders: 50 pending deliveries (Tagged: *Standard* or *Express*).
- Drivers: 10 available drivers.
- Costs: Fixed cost per driver per shift.
- External Variables: *Weather*: "Heavy Snow starting at 6:00 PM."
 - *Local Event*: "Major Football Match in the delivery zone causing congestion."

2. The Reasoning Engine The system must analyze the data and generate a decision. We are looking for your ability to identify:

- Operational Risk: (e.g., Impact of snow on delivery times).
- Financial Trade-off: (e.g., Cost of extra drivers vs. cost of missed SLAs).
- Actionable Prescription: A clear command for the manager.

3. Technical Execution

- Language: Python.
- Logic: You may use an LLM to assist with the reasoning, but we are evaluating the logic flow and decision clarity, not just the prompt.
- Output: A structured JSON object for a WhatsApp alert:

JSON

```
{
  "risk_level": "High",
  "risk_summary": "SLA failure likely due to weather + traffic",
  "estimated_impact": "Delivery speed reduced by approx 40%",
  "prescription": "Deploy 2 additional drivers OR prioritise express deliveries",
  "confidence_score": "85%"
}
```

4. Requirements

1. Codebase: A GitHub repository link.
2. Meeting discussion - google meet where both of you explain:
 - The logic behind the decision engine.
 - Division of Work: How you collaborated to build this as a team.

5. Timeline

- Duration: 72 Hours.
- Expectation: A working prototype that demonstrates "Product-Thinking," not a polished, production-ready app.